Beliz Kaleli

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Sr. Staff Security Research Engineer building AI-powered security systems for large-scale real-world threat detection in high-stakes environments. Proven track record of delivering high-performance production-grade systems that operate at scale; vulnerability research (CVEs, IOCs), TTP analysis, browser exploit identification and malicious infrastructure analysis. Primary inventor on 3+ patents, author of 2 award-winning research and 10+ high-impact threat intelligence publications. Core expertise includes designing autonomous AI agents for detection, with a focus on emerging attack vectors such as indirect prompt injections, vulnerability scanning, GenAI-driven phishing and malware delivery methods.

Work Experience

Sr. Staff Security Research Engineer | *Palo Alto Networks, Santa Clara, CA, USA* Oct 2024 – Present

- Led a cross-functional effort to design and develop a hybrid, multi-network vulnerability scanning detection system combining volume-based features, threat signatures, and ML to block ~1M malicious requests daily in real time (patented; first-author paper and blog).
- Built an AI-powered indirect prompt injection (IDPI) detection system protecting AI agents from web-based attacks; achieved first real-world detection of high-severity IDPI (AI reviewer bypass, data exfiltration, prompt leakage, and DoS) (patent pending; invited talk at Stanford University; first-author blog and post).
- Built an autonomous AI agent crawler to solve captchas, authenticate, and navigate complex page interactions to access previously inaccessible content (gated malicious downloads, phishing forms) critical for detection (placed 2nd among 30 teams at the Google Cloud/Palo Alto Networks Agentic AI Hackathon).
- Led a cross-team effort to design (PRD) and develop an AI-powered system for detecting OAuth flow abuse in real-time production traffic (patent pending; published 3 posts).
- Designed and implemented an agentic brand impersonation detection system using LLMs with RAG, multimodal analysis, and integrated tools to identify phishing attacks against global and regional brands.
- Primary inventor on 3+ patents (7+ in total) for AI-based security detection techniques, covering phishing, scams, indirect prompt injection, vulnerability scanning, and domain squatting attacks.
- Detected 30+ novel attack campaigns (IDPI, OAuth abuse, browser-in-the-middle, GenAI-based phishing), documented in 9 high-impact threat intelligence publications (6 as first author).

Staff Security Research Engineer | *Palo Alto Networks, Santa Clara, CA, USA* May 2023 – Oct 2024
Security Researcher Intern | *Palo Alto Networks, Santa Clara, CA, USA* Jun 2022 – Nov 2022
Software Engineer | *AYESAS, Ankara, Turkey* Sep 2017 – Jun 2018

Skills

Languages: Python, Go, SQL, C, C++, HTML/CSS, JavaScript, Typescript, Java
Tools & Frameworks: GCP, BigQuery, Vertex AI, Claude Code, Google ADK, VirusTotal, SIEM, MCP, Git, CI/CD, Docker, Kubernetes, Pub/Sub, Terraform, REST, gRPC, MongoDB, Scikit-Learn, LangChain, AWS

Awards & Honors

2nd Place at Hackathon | *Google Cloud/Palo Alto Networks Agentic AI Hackathon* 2026
Best Student Paper Award | *Security and Privacy in Communication Networks (SecureComm)* 2024
Best Student Paper Award Finalist | *The Center for Information & Systems Engineering (CISE)* 2020
Best Paper Award | *Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA)* 2019

Patents

- Agentic Web-Based Indirect Prompt Injection (IDPI) Detector Filed, Apr 2026
  ***B. Kaleli**, S. Farooqi, N. Mohamed, O. Starov*
- Detecting Scanning and Attacking URLs in Network Traffic Patented (US-12483573-B2), Nov 2025
  ***B. Kaleli**, F. Liu, O. Starov*
- Phantom Squatting: Anticipating GenAI-Hallucinated URLs Before Attackers Do Filed, Oct 2025
  *K. Nagaraj, D. Marty, **B. Kaleli**, O. Starov*
- Detecting Defensively Registered Domains Using LLMs Filed, Oct 2025
  *N. Mohamed, J. Chen, O. Starov, **B. Kaleli***
- AVA: Authentication Verification Analyzer for Detecting OAuth Flow Abuses Filed, Jun 2025
  ***B. Kaleli**, D. Marty, L. Hu, O. Starov*
- A Grounded AI-based Pipeline for Detecting Social Engineering Attacks Filed, Jun 2025
  *L. Hu, F. Liu, **B. Kaleli**, O. Starov*
- A Deep Learning Approach for Real-time Scanning URLs Detection Using IUPG Framework Filed, Feb 2025
  *T. Li, B. Kutt, **B. Kaleli**, F. Liu, P. Peng, O. Starov, W. Hewlett*

Academic Papers

- ScanWars: (A Multi-network Approach to Detecting and Analyzing) The Rise of Scanning Activity 2025
  ***B. Kaleli**, T. Li, F. Liu, O. Starov, M. Egele, G. Stringhini*
  Symposium on Electronic Crime Research (eCrime)
- EasyCSPeasy: A Server-side and Language-agnostic XSS Mitigation 2024
  ***B. Kaleli**, M. Egele, G. Stringhini*
  *Security and Privacy in Communication Networks (SecureComm) – **Best Paper Award***
- To Err.Is Human: Characterizing the Threat of Unintended URLs in Social Media 2021
  ***B. Kaleli**, B. Kondracki, M. Egele, G. Stringhini, N. Nikiforakis*
  Network and Distributed System Security Symposium (NDSS)
- Studying the Privacy Issues of the Incorrect Use of the Feature Policy 2020
  ***B. Kaleli**, M. Egele, G. Stringhini*
  Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb)
- On the Perils of Leaking Referrers in Online Collaboration Services 2019
  ***B. Kaleli**, M. Egele, G. Stringhini*
  *International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA) – **Best Paper Award***

Invited Talks & Service

Invited Speaker at Stanford University | Real World AI Security Conference 2026
Program Committee Member | Annual Computer Security Applications Conference (ACSAC) 2025
Invited Speaker at University of Michigan | Center for Computer Security and Society (SECRIT) 2021

Other Publications

Blogs

- Fooling AI Agents: Web-Based Indirect Prompt Injection Observed in the Wild Mar 2026
  ***B. Kaleli**, S. Farooqi, O. Starov, N. Mohamed*
- Fashionable Phishing Bait: GenAI on the Hook Aug 2025
  *Z. You, **B. Kaleli**, L. Hu, W. Wang*
- It Was Not Me! Malware-Initiated Vulnerability Scanning Is on the Rise Apr 2024
  ***B. Kaleli**, F. Liu, P. Peng, O. Starov, J. Allen, S. Springer*

Posts

- New Browser-in-the-Browser Phishing Campaign Jun 2026
  ***B. Kaleli**, G. Rane, O. Starov*
- Indirect Prompt Injection to Push a Malicious Chrome Extension May 2026
  ***B. Kaleli**, S. Farooqi, N. Mohamed, O. Starov*
- Phishing Campaign Pivots to Sophisticated Methods May 2026
  ***B. Kaleli**, O. Starov*
- Real-World Example of Malicious Indirect Prompt Injection Dec 2025
  ***B. Kaleli**, S. Farooqi, O. Starov*
- Crypto-themed Emails with Links to Sites Posing as Cryptocurrency Platforms Oct 2025
  ***B. Kaleli**, O. Starov*
- Phishing Campaigns Impersonate Major Business Platforms and Abuse OAuth Flows Oct 2025
  ***B. Kaleli**, O. Starov*
- Ongoing Phishing Campaign Using Browser-In-The-Middle Technique Sep 2025
  ***B. Kaleli**, O. Starov, L. Hu, S. Shuang, N. Mohamed, B. Duncan*

Education

Boston University *Boston, MA, USA*
Ph.D. & M.S., Computer Engineering, ECE | GPA: 3.8/4.0 Jan 2025

Middle East Technical University *Ankara, Turkey*
B.S., Electrical and Electronics Engineering Sep 2017